CAPITAL ONE: CLOUD MIGRATION TO AWS

Week 1 Case Study Summary | Cloud Computing

OVERVIEW

Capital One's move to Amazon Web Services (AWS) combined infrastructure migration with a broader change in how it develops and operates software. The company selected AWS as its predominant cloud infrastructure provider in 2016 and completed its exit from eight on-premises data centers in 2020. Published goals included faster provisioning, scaling with demand, improved resilience, and giving engineering teams more time to build customer-facing software. [1][2][4]

ARCHITECTURE

AWS identifies services including Amazon EC2 (compute), Amazon S3 (object storage), Amazon RDS (managed relational databases), and Amazon Connect. Its examples also describe AWS Lambda and Amazon DynamoDB. Later Capital One engineering articles describe a serverless-first approach using Lambda or containers with Amazon ECS and AWS Fargate. A mobile-app migration used separately deployed and scaled components and phased production traffic. These are documented patterns and examples, not one universal architecture for the whole bank. [1][3][4][6]

REPORTED OUTCOMES

AWS reports development-environment build time falling from three months to minutes, with new code released multiple times per day rather than monthly or quarterly. Disaster-recovery time in tests improved by 70%, and critical-incident resolution time and transaction errors each fell by 50%. The case study says 80% of nearly 2,000 cloud applications were built from the ground up. These are published case-study figures, not measurements from our own lab. [1]

COST AND LESSONS

A separate AWS case study reports 90% cost savings for *one application* migrated to Lambda; that is not an enterprise-wide migration saving. The reviewed sources do not disclose Capital One's total migration cost. For our lab, cost must be estimated separately using stated workload assumptions. The broader lesson is that migration also requires application modernization, staff training, secure governance, deployment automation, reliability testing, and ongoing cost controls. [1][5]

SOURCES

[1] AWS, “Capital One Completes Migration from Data Centers to AWS” (aws.amazon.com).

[2] Amazon/AWS, “Capital One Selects AWS as its Predominant Cloud Infrastructure Provider” (2016).

[3] Capital One Tech, “The serverless-first strategy experience”.

[4] Capital One Tech, “Embracing AWS Lambda and serverless architecture”.

[5] AWS, “Capital One Saves Developer Time and Reduces Costs by Going Serverless on AWS”.

[6] Capital One Tech, “How we migrated a customer-facing mobile app to AWS”.